

Protocol-Dependent Performance in Rotating Machinery Fault Diagnosis: Evidence from Sealed Compound Generalization

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Abstract

Reliable fault diagnosis in deployed rotating machinery requires models to generalize across physical units and compound-fault combinations. However, reported performance can be overstated when records from the same bearing enter both training and test sets or when sealed compound labels influence model selection. To isolate these effects, this study proposes the Sealed Compound Generalization Protocol (SCGP), which separates development, physical-unit-based nested model selection, and one-shot sealed confirmation without modifying the classifier. SCGP was applied to the HANOI HUST bearing archive using a 2×2 matrix of partitioning and endpoint aggregation choices. With record-grouped splitting, the fixed logistic-envelope reference achieved record-level AUROC/exact-set accuracy of 0.951/0.776 and bearing-aggregated values of 0.959/0.791. Bearing-grouped splitting reduced the corresponding values to 0.828/0.510 and 0.860/0.490. Nested selection within bearing-grouped outer folds further reduced them to 0.777/0.420 and 0.803/0.406, while the selected model configuration differed across folds. In a pre-access-frozen Paderborn extension containing 12 new physical bearing codes, inner-/outer-race AUROC was 0.344/0.719 and bearing-code-level exact-set accuracy was 3/12. Together, these results identify partitioning, aggregation, and selection as material determinants of apparent diagnostic performance. SCGP therefore provides a reproducible basis for treating source-domain scores as protocol-bound references in deployment-oriented validation.

Keywords: rotating machinery fault diagnosis; bearing fault diagnosis; compound fault diagnosis; evaluation protocol; physical-unit split

1. Introduction

Rolling-element bearings are critical to the safe and continuous operation of rotating machinery. Reliable condition monitoring should therefore support fault decisions before degradation causes unplanned downtime or secondary damage. In deployment, however, diagnostic performance must extend beyond records observed during development to new physical units, operating states, and fault compositions. This requirement is especially demanding for compound faults because detecting individual damaged components does not guarantee recovery of the complete fault set needed for maintenance decisions. The HUST bearing dataset supports studies under varied fault and operating conditions [1, 2], while a subsequent comparison showed that performance depends on both signal representation and learning model [3].

Recent research has addressed changing operating conditions mainly through transfer learning, domain adaptation, and domain generalization. A systematic review identifies transfer learning as a major approach to machinery diagnosis under distribution shift [4].

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Domain-generalization studies further seek models that retain diagnostic capability in unseen conditions without using target-domain labels. These advances improve representations and training objectives. Evaluation design then determines whether their reported gains are measured on independent, sealed evidence.

Evaluation design is therefore a separate source of uncertainty. If windows or records from the same bearing enter both training and test sets, the score can reflect repeated acquisition structure. It may therefore overstate transport to an unseen physical unit. Data leakage between vibration-signal partitions has indeed been shown to inflate bearing-diagnosis performance [5]. Vieira et al. subsequently formalized bearing-wise partitioning, multi-label fault prediction, and macro AUROC for more realistic bearing-diagnosis evaluation, including experiments on HUST compound faults and the Paderborn archive [6]. Bearing-wise splitting is now an established requirement. The unresolved problem is how to couple physical-unit independence with nested model selection, endpoint aggregation, and the timing of confirmation-label access in one auditable contract. Surveys of machinery domain generalization further report substantial heterogeneity in datasets, domain definitions, and validation protocols [7,8].

To address this gap, this study proposes the Sealed Compound Generalization Protocol (SCGP) as an auditable evaluation contract. SCGP separates exploratory development, physical-unit-based nested model selection, endpoint aggregation, and one-shot sealed compound confirmation without changing the classifier. The empirical audit evaluates three connected questions about the robustness of diagnostic evidence. First, how does performance change when partitioning moves from record groups to independent physical bearings? Second, does nested selection consistently favor the same model-representation configuration across bearing folds? Third, does a frozen configuration retain component-level discrimination and exact-set recovery after compound labels are opened once?

Accordingly, this study estimates protocol sensitivity within the registered HANOI HUST and Paderborn settings and defines the validation steps needed for broader cross-rig use.

The main contributions of this study are as follows:

1. SCGP defines a reproducible access contract for physical-unit independence, nested selection, endpoint aggregation, and one-shot opening of compound labels.
2. A crossed HANOI HUST audit quantifies changes across partition and aggregation choices, then examines selection stability across engineered and raw-waveform model families.
3. A documented HANOI HUST held-out confirmation, an initial three-code Paderborn pilot, and a disjoint pre-access-frozen 12-code Paderborn extension assess component-level and exact-set recovery under frozen configurations.

Figure 1 summarizes the access progression, and Table 2 formalizes the corresponding contract. The remainder of the paper is organized as follows. Section 2 positions SCGP relative to related diagnostic and evaluation methods. Section 3 describes the datasets, access states, model families, and statistical procedures. Sections 4 and 5 present the findings and their implications, respectively, and Section 6 concludes the study.

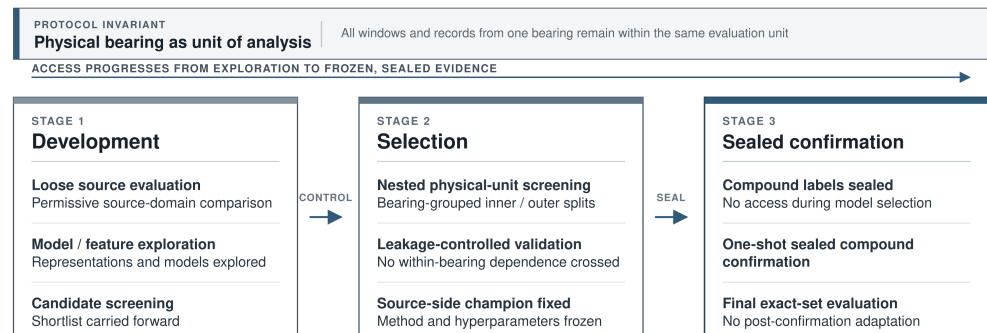


Figure 1. The SCGP access ladder. Evaluation progresses from exploratory source-domain development, through leakage-controlled physical-unit selection, to one-shot sealed compound confirmation. P0 and P1 are deliberately loose diagnostic states. From P2 onward, the physical bearing is the invariant unit: all windows and records derived from one bearing remain within the same evaluation partition.

2. Related Work

2.1. Diagnostic Models and Representations

A substantial body of work has focused on data-driven bearing fault diagnosis. Reviews synthesize convolutional architectures, representation-learning pipelines, and the broader development of deep models for mechanical diagnosis [9–11]. Comparative studies further show that measured performance depends on both the signal representation and learning model [12,13]. Representative systems include direct deep pipelines, lightweight multiscale networks, improved convolutional models, motor-bearing classifiers, and hierarchical feature learners [14–19]. Together, this literature establishes a broad candidate space. SCGP treats representation choice and physical-unit independence as complementary dimensions of diagnostic evidence.

2.2. Transfer Learning, Domain Adaptation, and Domain Generalization

Distribution shift across machines and operating conditions has motivated transfer learning and domain adaptation. These approaches transfer knowledge or align distributions using related source and target domains [4,20–26]. Their objectives include cross-machine transfer, multilayer alignment, benchmarked adaptation, and multisource partial adaptation. Domain generalization addresses a different access setting by learning from source domains without target-domain tuning. Existing strategies use conditional contrastive learning, meta-learning, agent-based objectives, and latent temporal representations [27–30]. Few-shot meta-learning and source-domain selection further address scarce labels and heterogeneous source candidates [31–33]. These methods primarily change the representation, objective, or training distribution. Their reported performance still depends on how physical units, selection data, and target labels are separated during evaluation.

2.3. Evaluation Dependence and Compound-Fault Endpoints

Model innovation and evaluation design address different sources of error. Bearing-specific dependence between vibration partitions can inflate diagnostic results when related windows or records cross the split boundary [5]. Surveys of domain generalization also identify heterogeneous datasets, domain definitions, and validation protocols as barriers to comparison [7,8]. A grouped split protects test independence only when every record and window from one bearing remains within one partition. Even then, selection can remain optimistic if held-out groups repeatedly influence representation, hyperparameter,

or threshold choices. SCGP therefore extends physical-unit separation by nesting model selection and keeping compound labels unavailable until confirmation.

The closest evaluation framework is the leakage-aware methodology of Vieira et al. [6]. That work established bearing-wise partitioning, a multi-label formulation, prevalence-independent component metrics, and experiments on co-occurring HUST faults. SCGP builds directly on these ingredients by joining physical-unit identity, inner-versus-outer selection permissions, record-versus-bearing aggregation, and confirmation-label opening in one machine-auditable access contract. The crossed audit then measures how each protocol choice changes the reported endpoint for a fixed diagnostic task.

Compound-fault diagnosis also creates a separate problem in defining evaluation endpoints. Existing approaches use blind deconvolution, adaptive filtering with kernel learning, EEMD–CNN pipelines, recurrence analysis, and spatiotemporal decomposition [34–38]. These methods provide technical routes for extracting or classifying compound signatures, while their output definitions and label-access conditions vary. Component-level discrimination can therefore coexist with an incorrect joint fault set. Evaluations should report component-level performance together with exact-set recovery to distinguish partial recognition from a correct compound decision.

2.4. Uncertainty-Aware Diagnosis and SCGP Positioning

Trustworthy diagnosis research has addressed predictive confidence through model calibration and uncertainty-aware learning [39–41]. Risk-controlled selective prediction provides a complementary mechanism for abstaining under a prescribed error risk [42]. SCGP adds the access controls needed to evaluate these confidence measures under physical-unit separation and fold-specific model selection.

Against this background, SCGP serves as an evaluation protocol that can accompany classifiers, adaptation objectives, compound-signal decomposition methods, and uncertainty estimators. It jointly specifies four elements: the independent physical unit, the model-selection boundary, the aggregation endpoint, and the one-shot compound-label opening. This positioning makes the evidential status of each score auditable as access moves from development to sealed confirmation.

3. Materials and Methods

3.1. Data Sources and Independent Experimental Units

The HANOI HUST bearing archive contains 99 MAT acquisitions from 33 physical bearings sampled at 51.2 kHz [1,2]. The source partition contains 57 acquisitions from 19 bearings, with three load records per bearing. These records represent normal, inner-race, outer-race, and ball conditions used for development. The remaining 42 acquisitions from 14 bearings form a documented held-out compound partition. Their numeric files were excluded during source development under a local access freeze. The retained provenance documents this freeze but contains no public pre-analysis timestamp, so the track is classified as historical held-out confirmation.

The physical bearing is the independent experimental unit, rather than a waveform window or acquisition record. All load records and derived windows from one bearing receive the same group identifier. Consequently, a held-out bearing contributes no waveform segment to its corresponding training fold. This rule controls dependence from repeated measurements of the same hardware and deterministic windows from the same acquisition.

Each waveform is converted into record-level features using a frozen windowing procedure. One-second windows contain 51,200 samples. Three extraction passes start at 0, 0.25, and 0.5 s, with non-overlapping windows within each pass. Features are computed for each technical window and summarized by their componentwise median and interquartile

range. This aggregation retains persistent spectral behavior while limiting the influence of isolated transients. The learning algorithms therefore operate on acquisition records, whereas inferential resampling remains at the bearing level.

The released source-window cache contains 1541 windows from the 57 source acquisitions. At 51.2 kHz, the offsets correspond to sample positions 0, 12,800, and 25,600. The cache stores the archive hash, window shape, bearing identifier, acquisition identifier, and feature-construction provenance. Compound numeric files remain unopened during source development. These fields permit an independent audit of the partition before model fitting.

Three external resources provide cross-rig observations at their native evaluation units. Each Mendeley Pump workbook represents one factorial state [43]. The Zenodo resource contributes a balanced 320-record mixture from one rig operated at four speeds [44]. Both are reported as supplementary quantitative checks. The initial Paderborn pilot contains 240 records nested within three bearing codes: K001, KA01, and KI01 [45,46]. Its rig, bearing type, sampling interface, and damage construction differ from HANOI HUST.

The initial Paderborn pilot contains one healthy code and two damaged codes. The source configuration, representation, threshold, and prediction procedure were frozen before external scoring. The component labels defined only the evaluation mapping and were not used for representation choice, hyperparameter tuning, threshold selection, or post-confirmation retraining. The effective independent unit is the bearing code.

To increase independent-unit support, a separate 12-code extension was registered before numeric access. It prespecified four healthy codes (K002–K005), four outer-race codes (KA04, KA15, KA16, and KA22), and four inner-race codes (KI04, KI14, KI16, and KI18). These codes are disjoint from the initial pilot and contribute 960 records in total. The same HANOI-only scaler, logistic-envelope model, $C = 10$, and threshold 0.5 were retained without Paderborn tuning. Archive hashes, official content lengths, the frozen script hash, and the 80-record-per-code structure were verified before scoring. The extension retains its own registry and access record, distinct from both the pilot and the earlier unscored 15-code registry.

3.2. Problem Formulation and SCGP Access Contract

Let $D_s = \{(x_i, \mathbf{y}_i, g_i)\}$ denote the source records, where g_i identifies the physical bearing. Each target $\mathbf{y}_i \in \{0, 1\}^3$ encodes inner-race, outer-race, and ball faults. Let $\mathcal{C}$ be the registered candidate set and D_c the confirmation records whose compound labels remain unavailable during development. We use I_0 to denote the source-only information budget: model fitting and selection may use D_s and registered source metadata but no numeric values or labels from D_c or an external boundary set. A separate binary estimator returns the positive-class probability for each component. A fixed threshold of 0.5 converts each probability into a component decision. Exact-set correctness requires all three decisions to match the target vector.

SCGP governs three chronological access phases. Development permits source data for representation and candidate comparison. Selection fixes the candidate and hyperparameters through a nested physical-unit procedure. Confirmation keeps compound labels sealed until one-shot evaluation and prohibits post-opening tuning. Thus, both the executable code and the permitted information flow are frozen.

These phases are instantiated as five evaluation states. P0 uses a window-random partition and is retained only as a descriptive leakage diagnostic. P1 uses a record-grouped split in which windows from one acquisition remain together, while records from the same bearing may cross partitions. P2 groups by physical bearing, P3 adds nested model

Table 1. SCGP access contract. Each state is defined before source or compound results are interpreted.

State	Partition rule	Unit of analysis	Label access	Role in the analysis
P0	Window-random split	Window	Open	Descriptive leakage diagnostic; excluded from primary inference
P1	Record-grouped split	Record	Open	Windows stay within records, but records from one bearing may cross partitions
P2	Physical-unit group split	Bearing	Open	Leakage-controlled source-side comparison
P3	Nested physical-unit selection	Bearing	Open during selection, frozen before confirmation	Outer evaluation of source-side model selection
P4	Sealed compound confirmation	Bearing	Sealed until one-shot confirmation	Final confirmation state for compound recovery and boundary checks

Table 2. Positioning SCGP relative to commonly used evaluation practices. Its operational contribution is an explicit, auditable contract linking physical-unit identity, selection permissions, compound-label opening, and joint endpoints.

Existing practice	Typical guarantee	SCGP-specific operational addition
Grouped cross-validation	Reduces same-group leakage	Declares the physical bearing as the independent unit and records its split manifest and aggregation unit
Nested cross-validation	Separates tuning from outer evaluation	Freezes the candidate, preprocessing, threshold, and selection permission at an explicit access boundary
Locked or held-out test set	Protects a final score from tuning	Seals compound labels until predictions and audit hashes are written, with one-shot opening recorded
Multilabel metrics	Reports component or set performance	Requires component-level and exact-set endpoints at the same declared unit, with per-unit uncertainty

selection, and P4 seals compound labels until confirmation. Table 1 defines the permitted data access and analysis unit. Results from different states are not interchangeable.

For each outer split, SCGP first partitions D_s by the declared unit. Every representation and candidate is fitted only on permitted inner-training records. The prespecified endpoint selects c^* , which is refitted on the allowed development records and evaluated once on untouched outer records. In a prospective P4 implementation, the frozen c^* , representation, preprocessing parameters, and threshold are transferred unchanged to D_c . Compound labels are opened only after predictions and audit hashes have been written. Confirmation results cannot alter the candidate, feature map, or threshold. Each run outputs its split manifest, selected candidate, predictions, unit-level metrics, and access log. The empirical tracks below are labeled according to the strongest access history their retained evidence supports.

Table 2 distinguishes this operational contract from its individual ingredients. SCGP operationalizes unit identity, selection permissions, label opening, and joint endpoints within one auditable protocol.

3.3. Signal Representations

The frozen representation family contains robust time statistics, fixed physical-frequency log-power bands, and envelope log-power bands. The statistics block includes the mean, standard deviation, root-mean-square value, absolute mean, extrema, and peak-to-peak range. It also includes skewness, excess kurtosis, crest factor, impulse factor, shape factor, and an envelope-based peak factor. Median and interquartile-range aggregation produces 28 values per extraction offset and 84 values after the three offsets are concatenated.

The two spectral views use the same frequency partition, isolating representation type rather than dimensionality. After mean removal and Hann tapering, squared real-

Fourier magnitudes are summed within 64 equal-width bands from 0 to 25.6 kHz. A base-10 logarithm is applied after a fixed numerical floor. The envelope view takes the Hilbert-transform analytic magnitude and removes its mean. The same taper, transform, band integration, and logarithm are then applied. Median–interquartile aggregation yields 128 values per offset and 384 values for either spectral view. The all representation concatenates the statistics and both spectral blocks.

The supplementary physics-feature variant maps fixed and envelope log-power spectra to an order axis using recorded shaft speed. Linear interpolation produces 64 bins spanning 0.5–256 orders. Missing or non-positive speed produces an all-zero 64-bin view. Speed was available for 24 of the 57 HANOI source records and missing for 33; no missingness indicator was appended. This zero-fill convention defines a supplementary comparator within HANOI source folds, with no cross-rig physical equivalence assumed.

3.4. Candidate Models and Nested Selection

The prespecified source-side ranking endpoint is mean component AUROC. It selected an L2-regularized logistic model with `envelope_log_power`, which was frozen before confirmation. Features are standardized within each training fold, class weights are balanced, $C = 10$, and optimization is capped at 5000 iterations. The threshold remains 0.5 and is not optimized on the compound partition. Threshold sensitivity is treated as an operating-boundary analysis, not another selection step.

The registered source comparison also includes logistic models, linear and radial-basis SVMs, random forests, extremely randomized trees, an empirical-prior control, multilayer perceptrons, and implemented NOACE variants. Compact tree and kernel grids test protocol sensitivity across distinct decision rules without unrestricted search. The comparator set is layered by evidential role. The source reference and representation ablation form the pre-specified internal comparison. Classical and modern alternatives test whether the observed protocol pattern depends on an unusually weak estimator. Confirmation-stage models remain separate because they answer a different question from source-side selection.

For provenance, the machine-readable files use two audit identifiers. G2 denotes the primary crossed partition–aggregation analysis, whereas G3 denotes a secondary source-only model-family audit. G3 adds random forest, source-only GroupDRO-style logistic reweighting, MiniROCKET, WDCNN, ResNet1D, and InceptionTime-style comparators. These models use only bearing-grouped outer splits, and their results leave the primary model choice unchanged. The supplementary physics representation is distinct from the MLP comparator identified as `mlp_deep`.

NOACE is the implemented nuisance-residualized compositional-evidence comparator. Here, “orthogonal” means numerical orthogonality to the fitted training design, not causal nuisance removal. Within each training fold, a linear nuisance model regresses the feature vector on one-hot bearing-type and load indicators. The residual is passed to the downstream estimator. Nuisance coefficients and category levels are frozen before test prediction. If a bearing type is absent from the training categories, the implementation raises an error rather than extrapolating silently. Bearing type is therefore treated as available metadata, not a universally available sensor measurement.

Let $\mu_{\emptyset}$ be the healthy residual mean and μ_j the singleton mean for component $j \in \{1, 2, 3\}$. For subset S , the additive prototype is $\mu_S = \mu_{\emptyset} + \sum_{j \in S} (\mu_j - \mu_{\emptyset})$. With diagonal residual variance v_k and subset prototype μ_S , a record with residual feature vector r receives subset energy $E_S(r) = \frac{1}{2} \sum_k (r_k - \mu_{S,k})^2 / v_k$. Subset weights are proportional to $\exp[-E_S(r)/T]$ with fixed temperature $T = 1$ and equal subset priors; component probabilities are obtained by marginalizing subset indicators. The registered ablations vary

residualization, additive prototypes, subset prior, temperature, and diagonal versus pooled variance.

The exact candidate identifiers, parameter values, split manifests, and prediction targets are serialized in the supplied result files. Remaining classical candidates use registered compact grids rather than unrestricted search. G3 comparators share the bearing-grouped outer splits, physical-bearing aggregation, and the I0 source-only information budget. G3 remains a separate source-only audit of model-family sensitivity.

3.5. Implementation and Reproducibility

The analysis was executed with Python 3.11. Waveform models use PyTorch 2.6, torchvision 0.21, torchaudio 2.6, and CUDA 12.4. NumPy, SciPy, pandas, and scikit-learn support feature processing and classical models. The recorded audit environment reports NumPy 2.3.5, SciPy 1.16.3, and scikit-learn 1.8.0. Classical models use frozen feature caches. Neural baselines use the recorded GPU implementation and fixed downsampling and training budgets.

The accompanying reproducibility package includes source builders, feature caches, split manifests, archive hashes, prediction arrays, and audit JSON files. The crossed endpoint audit contains the three record-prediction arrays, `factorial_endpoint_metrics.json`, `hanoi_hust_noace_ablation.json`, and the executable reconstruction scripts. The package also contains separate Paderborn registries for the three-code pilot and 12-code extension, frozen scoring scripts, derived feature caches, unit-level predictions, and an executable extension audit.

Randomness is separated by role. The frozen HANOI confirmation uses the registered random state 20260727, the primary bearing-cluster bootstrap uses seed 20260728 with 10,000 resamples, and the source stability audit repeats the fixed protocol with seeds 42, 7, 21, 84, and 168. The Paderborn extension uses seed 20260802 for 10,000 stratified bearing-code bootstrap resamples. These repetitions quantify uncertainty or stability under the registered endpoint. Every prediction file is linked to its split manifest, model family, representation, seed, and source-code provenance.

3.6. Evaluation Metrics, Aggregation, and Inference

Both component- and set-level metrics are reported because one accuracy value cannot characterize compound recognition. Component metrics are AUROC, balanced accuracy, AUPR, Brier score, and macro-F1. Set metrics are exact-set accuracy and Hamming loss. Boundary analyses additionally report calibration error, selective accuracy, risk–coverage behavior, and efficiency.

Let $\mathbf{y}_i \in \{0,1\}^3$ denote the true component vector for record i , $\hat{\mathbf{y}}_i$ the thresholded prediction, and $\mathbf{p}_i$ the corresponding probability vector. Exact-set accuracy is

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}(\mathbf{y}_i = \hat{\mathbf{y}}_i),$$

whereas Hamming loss averages disagreements over the $3n$ binary decisions. Component metrics are calculated separately for the three faults and then averaged. Favorable marginal rankings can therefore coexist with an incorrect joint label set.

Each reported endpoint names its analysis unit. G2 record-level values are calculated after window-to-record aggregation, so each acquisition contributes one prediction. For bearing-aggregated values, the three load-record probabilities are averaged within each physical bearing before metric calculation. G3 and the legacy NOACE comparisons use bearing-level endpoints after the same load aggregation. Exact-set accuracy always eval-

uates the joint vector at the stated unit. Confidence intervals resample physical bearings, never records or windows.

The primary crossed comparison varies partitioning (record-grouped versus bearing-grouped) and endpoint aggregation (record level versus bearing aggregated) for the fixed reference, forming the stated 2×2 matrix. The nested bearing-grouped cell adds a separate model-selection layer. The audit stores pre-aggregation record predictions and post-aggregation arrays for all reported cells, together with split manifests and the aggregation script. Within each fixed split cell, record-level and bearing-aggregated summaries reuse the same fitted-model outputs, preprocessing, class weighting, and threshold 0.5. Only prediction-stage aggregation and endpoint unit differ within that cell. The nested cell instead uses the candidate selected within each outer fold.

For metric m_s in split s , the reported point estimate is the registered repeated-split mean $\bar{m} = S^{-1} \sum_{s=1}^S m_s$. Displayed prediction counts are descriptive repeated-split counts, not independent-sample proportions. Exact-set intervals are bearing-cluster bootstrap intervals computed from per-bearing mean outcomes and are conditional on the registered repeated-split procedure. When all observed bearing outcomes are identical, this empirical bootstrap is degenerate; an exact binomial interval is then reported as a sensitivity analysis for population uncertainty.

For the 12-code Paderborn extension, records are first averaged within bearing code. Exact-set uncertainty uses a two-sided 95% Clopper–Pearson interval over 12 binary code outcomes. Component-AUROC intervals use 10,000 stratified bearing-code bootstrap resamples that preserve the positive and negative code counts for each component. These intervals characterize uncertainty over the observed external code set.

For the nested cell, model identity is fold-specific rather than globally frozen. Inner selection is completed before outer-test prediction. One selected candidate is refitted per outer fold and produces every test-record probability in that fold. Bearing aggregation is the deterministic mean of its three load probabilities. Probabilities from different candidates are never mixed, and every candidate uses the 0.5 threshold.

Calibration is evaluated at component and set levels. Component expected calibration error compares probabilities with empirical binary outcomes within confidence bins. Set confidence is the model-implied confidence assigned to the complete thresholded vector. Set-level calibration error compares this value with exact-set correctness. Selective prediction orders records by set confidence and reports exact-set accuracy as coverage decreases. The area under the risk–coverage curve summarizes the trade-off. Individual coverage points are retained because non-monotonic curves can reveal high-confidence failure groups.

Comparisons are made at the physical-bearing level. Percentile 95% confidence intervals use 10,000 bearing-cluster bootstrap resamples with seed 20260728. Registered paired tests use exact-set correctness after the three load probabilities are averaged. Each bearing therefore contributes one binary paired value. Because the support is discrete and ties are frequent, the Wilcoxon signed-rank result is retained only as a supplementary descriptive check. The main evidence is the per-bearing difference, its cluster-bootstrap interval, and the complete outcome table.

The 100 G3 outer splits are correlated resamples of the same 19 bearings. Split-level sign-permutation values are therefore conditional resampling summaries, not evidence from 100 independent experiments. Bearing-level bootstrap intervals and per-bearing outcomes define the inferential unit.

Outer-fold selection frequencies characterize stability without merging access states into a common leaderboard. They indicate whether the nested procedure repeatedly prefers one model–representation configuration.

Table 3. Primary G2 protocol comparison. Values are means across 100 registered splits. Record-level and bearing-aggregated endpoints within each fixed cell reuse the same fitted-model predictions; the nested row uses the candidate selected independently within each outer fold.

Protocol cell	Record AUROC	Record exact-set	Bearing AUROC	Bearing exact-set
Record-grouped, fixed reference	0.951	0.776	0.959	0.791
Bearing-grouped, fixed reference	0.828	0.510	0.860	0.490
Bearing-grouped, nested selection	0.777	0.420	0.803	0.406

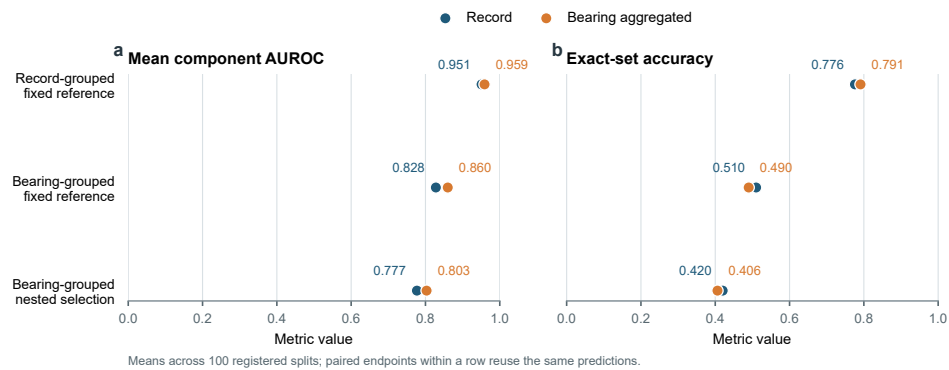


Figure 2. Primary matched-protocol results across 100 registered splits. (a) Mean component AUROC and (b) exact-set accuracy for record-grouped fixed-reference, bearing-grouped fixed-reference, and bearing-grouped nested-selection cells. Blue points use record-level endpoints; orange points aggregate the same outer-test probabilities by physical bearing. Lines join endpoints within a protocol cell and do not represent trajectories. All axes span 0–1. The nested candidate is selected independently inside each outer fold. Intervals and outcome counts are reported in the text because the repeated splits reuse the same 19 bearings and are not independent experiments.

4. Results

Results are reported in two documented analysis generations. The crossed G2 audit provides the primary matched-protocol estimates. The earlier 19-bearing source and NOACE audit is retained for traceability and is identified as the legacy audit throughout. Each generation retains its original access state and endpoint definition.

4.1. Crossed Partition–Aggregation Analysis

Table 3 gives the central protocol comparison. The first two rows form the fixed-reference 2 × 2 partition–aggregation matrix. The third row adds nested selection within bearing-grouped outer folds.

The crossed audit retained record-level predictions and bearing-aggregated predictions for each protocol cell. With record-grouped splitting, record-level AUROC was 0.951 (95% CI, 0.864–0.987). Exact-set accuracy was 0.776 (1086/1400; 95% CI, 0.626–0.899). After bearing aggregation, AUROC was 0.959 (0.884–0.993), and exact-set accuracy was 0.791 (866/1093; 0.635–0.919).

With fixed bearing-grouped splitting, record-level AUROC was 0.828 (95% CI, 0.629–0.902). Exact-set accuracy was 0.510 (765/1500; 95% CI, 0.417–0.747). Bearing aggregation produced AUROC 0.860 (0.656–0.927) and exact-set accuracy 0.490 (245/500; 0.400–0.776).

Nested bearing-grouped selection produced record-level AUROC 0.777 (95% CI, 0.559–0.846) and exact-set accuracy 0.420 (630/1500; 0.345–0.686). The bearing-aggregated values were 0.803 (0.567–0.850) and 0.406 (203/500; 0.328–0.701), respectively.

Across the 100 nested outer folds, logistic-envelope configurations were selected 54 times. Extra-trees-envelope and RBF-SVM-envelope configurations were selected 29 and 17 times, respectively. All selected configurations used the registered envelope view.

Exact hyperparameter frequencies are provided in the split manifest. The nested estimates are therefore fold-specific selection results rather than fixed-reference estimates. Figure 2 makes the central pattern explicit: physical-unit separation reduced both endpoints, nested selection reduced them further, and bearing aggregation did not recover the looser record-grouped performance.

4.2. Secondary Legacy Source Reference and Representation Dependence

On the 19-bearing source partition, the prespecified mean component AUROC endpoint selected `logistic_12 + envelope_log_power` (AUROC, 0.814). Its balanced accuracy was 0.743, AUPR 0.661, Brier score 0.161, macro-F1 0.743, exact-set accuracy 0.544, and Hamming loss 0.199. With the same classifier, the `a11` representation reduced AUROC by 0.007 but increased exact-set accuracy by 0.053 and reduced Brier score by 0.020. The full endpoint-dependent ranking and representation profile are retained in Supplementary Figure S1 and Supplementary Data S1 rather than interleaved with the primary matched-protocol sequence.

4.3. Repeatability and Negative Controls

All five registered seeds (42, 7, 21, 84, and 168) selected the legacy logistic-envelope configuration. Their aggregate results were identical, and no seed-dependent variation was observed.

After label shuffling, mean component AUROC ranged from 0.325 to 0.565 across five seeds. Exact-set accuracy ranged from 0.175 to 0.316. Both ranges were below the unshuffled values of 0.814 and 0.544.

Identity probes quantified two potential nuisance signals. Bearing type remained predictable under load-held-out evaluation, with balanced accuracy 0.872. Load prediction under bearing-held-out evaluation produced balanced accuracy 0.281.

The source partition contained five bearing-type codes (4–8). Type 4 had no ball-state record. Types 5–8 contained normal, inner-race, outer-race, and ball records across three loads. A leave-type-out experiment was not performed, so hardware-type and label dependence remained unresolved.

4.4. Held-Out Compound and Cross-Rig Boundary Checks

The documented HANOI HUST held-out confirmation contained 14 compound bearings. After bearing aggregation, `noace_classical` achieved mean component AUROC 0.621 (95% CI, 0.496–0.739). Its exact-set accuracy was 0/14, with a degenerate empirical cluster-bootstrap interval of 0–0. The fixed logistic-envelope reference achieved AUROC 0.811 (0.668–0.923) and exact-set accuracy 2/14, or 0.143 (0–0.357). The MLP comparator, `mlp_deep`, achieved AUROC 0.881 (0.706–1.000) and exact-set accuracy 0/14, again with an empirical interval of 0–0. For either 0/14 outcome, the two-sided exact binomial sensitivity interval was 0–0.232.

The Mendeley Pump and Zenodo workbook-level analyses used task-specific algorithms and evaluation units. Their results are reported in Supplementary Information as additional cross-rig observations.

The Paderborn one-shot pilot contained 240 records from three physical bearing codes. The fixed logistic-envelope reference correctly identified the healthy code but missed both damaged codes. Component AUROC was 0.000 for the inner- and outer-race components, and bearing-code-level exact-set accuracy was 1/3. Scores were aggregated across the three bearing codes, and no external tuning was performed.

The disjoint Paderborn extension contained 960 records from 12 physical bearing codes. Inner-race AUROC was 0.344 (stratified bearing-code bootstrap 95% CI, 0–0.688), and outer-race AUROC was 0.719 (0.250–1.000). Exact-set accuracy was 3/12, or 0.250

Table 4. Documented HANOI HUST held-out confirmation and Paderborn cross-rig boundary checks.

Resource	Evaluation unit	Observed result	Evidential interpretation
HANOI HUST	14 held-out compound bearings	Across three models, component AUROC was 0.621–0.881 and exact-set accuracy was 0–0.143.	Historical held-out confirmation under a documented local freeze.
Paderborn pilot	3 physical bearing codes (240 records)	Inner-/outer-race AUROC was 0.000/0.000; exact-set accuracy was 1/3, with only the healthy code recovered.	Frozen one-shot score; negative pilot evidence from three independent codes.
Paderborn extension	12 new physical bearing codes (960 records)	Inner-/outer-race AUROC was 0.344/0.719; exact-set accuracy was 3/12 (95% exact CI, 0.055–0.572).	Pre-access-frozen, disjoint external score without Paderborn tuning across 12 new codes.

Mendeley Pump and Zenodo single-rig checks are reported in Supplementary Information at their native tasks and evaluation units.

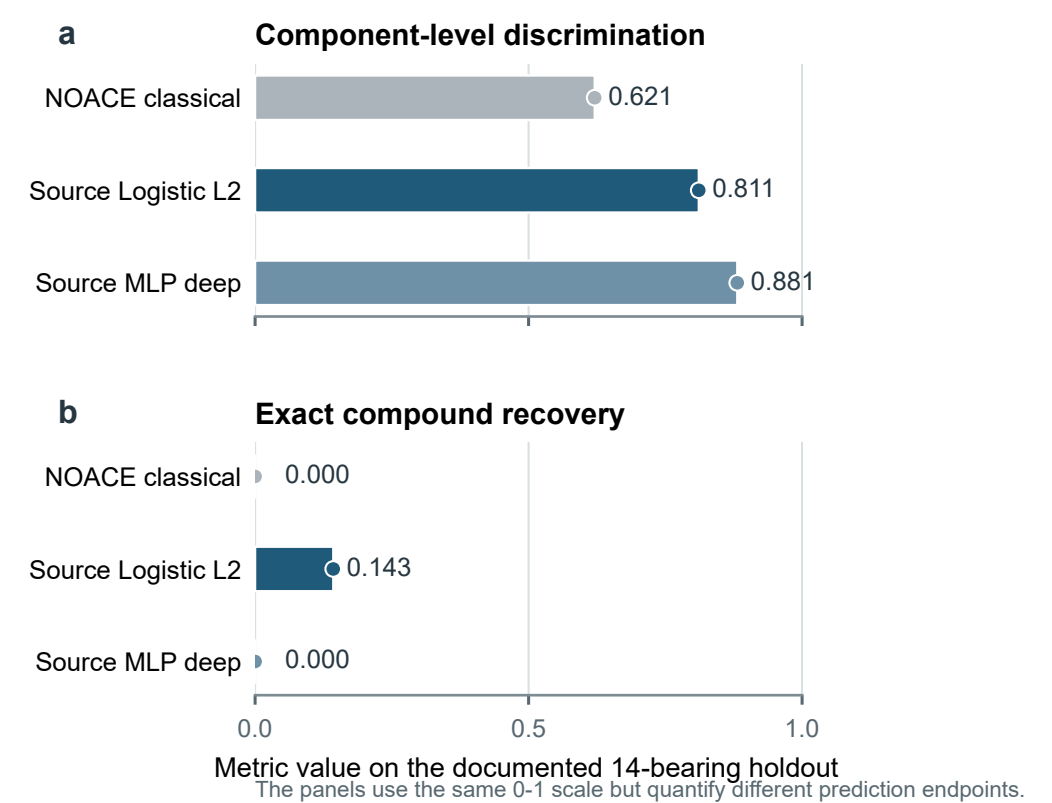


Figure 3. Documented held-out confirmation on 14 compound bearings. (a) Mean component AUROC and (b) exact-set accuracy for three models. Both panels use a 0–1 scale but represent different endpoints and are not directly subtractable. Bars are point estimates; the small 14-bearing boundary produces wide or degenerate discrete intervals, which are reported in the Results text and audit file rather than overlaid on the compact plot. Component AUROC ranges from 0.621 to 0.881, whereas exact-set accuracy ranges from 0 to 0.143.

(95% exact CI, 0.055–0.572), with correct recovery for 2/4 healthy, 0/4 outer-race, and 1/4 inner-race codes. The frozen source model and threshold generated these results under the extension’s registered access record.

The documented held-out and external boundary tracks are summarized in [Table 4](#). [Figure 3](#) contrasts component and exact-set endpoints on HANOI HUST.

Table 5. Source-only G3 robustness audit under the identical 100 bearing-grouped outer splits. All scores are computed after aggregation to the physical-bearing unit. The raw-waveform models use a fixed downsampling factor of 64 and are reported as diagnostic R2 implementations; no model was tuned on the outer test bearings.

Method	Input / budget	AUROC	AUPR	Bal. Acc.	Macro-F1	Exact set
Fixed logistic	Envelope features / I0	0.862	0.835	0.742	0.704	0.490
Random forest	Envelope features / I0	0.785	0.750	0.666	0.628	0.458
GroupDRO-style logistic	Envelope features / I0	0.845	0.825	0.721	0.680	0.482
MiniROCKET	Raw waveform / I0	0.751	0.708	0.682	0.628	0.556
WDCNN-style	Raw waveform / I0	0.630	0.608	0.530	0.386	0.116
ResNet1D	Raw waveform / I0	0.609	0.607	0.521	0.373	0.120
InceptionTime-style	Raw waveform / I0	0.569	0.565	0.516	0.370	0.120

4.5. Secondary Nested-Selection and Model-Family Audits

The legacy 19-fold nested physical-unit screen achieved mean component AUROC 0.701 and balanced accuracy 0.685. Exact-set accuracy was 0.526, and the Brier score was 0.169. Relative to its frozen source evaluation, AUROC decreased by 0.112, balanced accuracy by 0.058, and exact-set accuracy by 0.018. Candidate pools and endpoint targets differed between these procedures.

The exact-set endpoint ranked random forest with statistics first at 0.579. Extra trees with envelope log power reached 0.544. Linear SVM with envelope log power and RBF-SVM with all features had lower mean component AUROC than the legacy source reference.

Across 19 leave-one-bearing-out folds, `logistic_l2 + all` accounted for 10 selections after hyperparameters were pooled. `logistic_l2 + statistics` accounted for four selections. The two `extra_trees` views accounted for four, and `logistic_l2 + fixed_log_power` accounted for one.

This histogram uses a different candidate pool and endpoint from the primary 100-fold audit. Its counts are therefore not directly comparable with the primary 54/29/17 selection frequencies.

The G3 audit compared seven model families under identical 100 bearing-grouped outer splits and bearing aggregation. All models used I0 source-only access. Comparators were random forest, GroupDRO-style logistic regression, MiniROCKET, WDCNN, ResNet1D, and InceptionTime.

As shown in Table 5, the fixed logistic-envelope reference had the highest AUROC, 0.862. GroupDRO-style logistic regression reached 0.845, a difference of 0.017. Among raw-waveform models, MiniROCKET had the highest AUROC (0.751) and exact-set accuracy (0.556). WDCNN, ResNet1D, and InceptionTime produced AUROC values of 0.630, 0.609, and 0.569, respectively. The split-level sign-permutation summary is retained in Supplementary Information because all 100 splits reused the same 19 bearings.

The corresponding legacy score changes and foldwise selection frequencies are shown in Supplementary Figure S2. They remain secondary because the candidate pool and target endpoint differ from those of the primary G2 audit.

4.6. Secondary Comparator Uncertainty and NOACE Ablation

The complete legacy comparator table and paired statistics are reported in Supplementary Table S4 rather than repeated in the main evidence sequence.

The legacy source exact-set accuracy of 0.544 had a 95% bearing-cluster bootstrap interval of 0.333–0.754. The paired difference from `NOACE-classical` and `NOACE-physics` was −3.51 percentage points (95% CI, −28.1 to 21.1). The Holm-adjusted *p* value was 1.000, and *d_z* was −0.061. The difference from the MLP comparator was 0.00 percentage points (95% CI, −17.5 to 15.8; Holm-adjusted *p* = 1.000).

Across the same 19 leave-one-bearing-out folds, the full residualized additive NOACE model reached AUROC 0.812. Exact-set accuracy was 0.579 (95% CI, 0.368–0.789). Removing residualization increased the point estimates to 0.823 and 0.632 (0.421–0.842), respectively.

Replacing additive subset prototypes with a healthy-only prototype reduced AUROC to 0.500 and exact-set accuracy to 0 (95% CI, 0–0). Pooled variance reduced AUROC to 0.646 and exact-set accuracy to 0.474 (0.263–0.684). Temperatures $T = 0.5$ and $T = 2$ produced AUROC values of 0.805 and 0.807, respectively. Exact-set accuracy remained 0.579 (0.368–0.789) at both temperatures. The source-frequency subset prior was unchanged at the displayed precision.

Foldwise residual–nuisance correlations were below 3.3×10^{-15} . This value follows algebraically from least-squares residualization against the fitted training design and is not an independent performance result.

4.7. Secondary Calibration, Stress, and Efficiency Audits

Set-level expected calibration error was 0.298, and the area under the risk–coverage curve was 0.309. Component expected calibration error was 0.169 for the inner race, 0.296 for the outer race, and 0.068 for the ball component.

Across component thresholds from 0.2 to 0.8, record-level exact-set accuracy ranged from 0.491 to 0.561. Bearing-level exact-set accuracy ranged from 0.474 to 0.579. This prespecified scan was performed after frozen predictions were generated. Component-specific and joint exact-set-optimized thresholds were not estimated. The headline endpoint retained the prespecified threshold of 0.5.

At full coverage, selective exact-set accuracy was 0.544. Retaining the 21% most confident records increased accuracy to 0.917 at a confidence threshold of 0.982. At 32% coverage, accuracy was 0.778. Across intermediate coverage levels, it ranged from 0.625 to 0.793, so the risk–coverage curve was non-monotonic.

Bearings B5, B6, I7, O4, and O5 had zero exact-set accuracy across their three records. Their mean confidence ranged from 0.756 to 0.928. Values reached 0.928 for B5 and 0.888 for O5, documenting high-confidence group failures.

Waveform perturbations were applied before feature extraction. At 10 dB additive white noise, mean component AUROC was 0.805 and exact-set accuracy was 0.509. At −5 dB, these values were 0.589 and 0.316. Impulsive corruption at rate 0.10 reduced exact-set accuracy to 0.281. Clipping to 60% of the original amplitude range reduced AUROC to 0.654. With 30% sample dropout, AUROC was 0.731 and exact-set accuracy was 0.544. A 20% circular shift produced AUROC 0.831 and exact-set accuracy 0.596.

Under −5 dB additive noise, NOACE-classical retained AUROC 0.784 and exact-set accuracy 0.579. Under the same corruption, the fixed logistic–envelope reference reached 0.589 and 0.316. The clean Brier scores were 0.212 for NOACE-classical and 0.161 for the fixed reference.

Median evaluation time over three repeats was 0.129 s for NOACE-classical, 1.890 s for the fixed reference, and 8.791 s for the MLP comparator. The corresponding 95th-percentile times were 0.148, 1.895, and 9.457 s. Python-level peak allocations were 2.11, 3.23, and 0.62 MiB, respectively. These values describe the recorded implementation and hardware environment.

5. Discussion

5.1. Interpreting Protocol-Dependent Performance

The crossed audit separates four reporting dimensions that a single diagnostic score would otherwise conflate. Aggregation captures the change between record-level and

bearing-aggregated endpoints within one split state. Its effect was endpoint-specific because the direction differed across metrics. Partitioning captures the difference between record-grouped and bearing-grouped evaluation at a fixed endpoint. Selection captures the additional change when candidate choice enters the outer boundary. The output dimension distinguishes component discrimination from recovery of the complete fault set.

The crossed design identifies which reporting choice changed and which information remained available. It therefore extends concerns about vibration-signal leakage and heterogeneous domain definitions in machinery diagnosis [5,7,8]. Because candidate pools and endpoint targets differed in some states, the observed changes are descriptive protocol contrasts rather than additive causal effects.

SCGP complements transfer learning and domain generalization by controlling evaluation access. Those methods change representations, training objectives, or available domains [4,7,20,27]. SCGP records physical-unit independence and prevents confirmation labels from influencing selection. Building on the leakage-aware framework of Vieira et al. [6], it integrates outer-unit independence, nested selection, endpoint aggregation, and confirmation access within one traceable protocol.

5.2. Representation, Confidence, and External Transport

Envelope log power was a conditional source reference because it led the prespecified component-AUROC endpoint. Its perturbation profile was compatible with a modulation-energy interpretation but did not establish that mechanism. Time shifts were tolerated, whereas impulsive corruption and clipping were damaging. Bearing type was also strongly encoded. Part of the source advantage may therefore reflect hardware-specific structure rather than a transportable fault signature.

Bearing-grouped evaluation removed direct reuse of the same hardware, while residual type predictability indicated remaining confounding. Seed stability and label shuffling supported the internal feature-label association. Held-out and external confirmation then tested transport to new physical units, rigs, and compound compositions.

Calibration results further show why ranking alone is insufficient for maintenance decisions. High-confidence group failures and a non-monotonic risk-coverage curve indicate that confidence did not consistently order errors. Calibration and selective prediction characterize confidence after the evaluation population is defined [39–42]. A deployment rejection policy would additionally require validation by physical group and operating state.

The external tracks contribute distinct observations across heterogeneous rigs and evaluation units. Mendeley and Zenodo provide supplementary single-rig checks. The three-code Paderborn pilot supplied initial negative evidence, and the disjoint 12-code extension showed that poor exact-set transport persisted across more physical units. Together, these results reinforce the difference between extracting component signatures and recovering a complete compound label set [34–38]. Their separate access histories and sampling designs support separate reporting rather than a pooled population estimate.

5.3. Limitations and Evaluation Priorities

Several limitations bound the present inference. The source and held-out HANOI HUST analyses contain only 19 and 14 physical bearings, respectively. Their cluster-bootstrap intervals are consequently wide, and a few difficult bearings can materially change exact-set accuracy. The source and confirmation-stage candidate sets also differ at some access states. Selection frequencies therefore document stability within each registered state.

The source labels primarily represent normal and single-component states, whereas confirmation concerns compound combinations. The exact-set gap combines distribution change with a more difficult label-composition problem. It cannot be attributed to the access protocol alone. Separating these effects requires a factorial compound boundary on matched physical units or an independently registered task-matched control.

External resources were sampled from limited rigs or factorial states rather than a defined deployment population. The Paderborn extension increases the external support from three to 12 new bearing codes, but it still contains only four codes per state from one rig and no Paderborn ball-fault state. The synthetic waveform perturbations are similarly controlled stresses. They reveal sensitivity to specified corruptions but do not reproduce every change in sensor, mounting, load, or environment encountered in service.

SCGP makes these boundaries auditable by recording available evidence, independent physical units, prediction aggregation, selection permissions, and the timing of label opening. A stronger replication should include more independent units, one candidate grid across P2–P4, and an independently registered compound boundary.

For applied signal processing, reproducibility requires recoverable code, partitions, selection permissions, independent units, and confirmation events. Deployment-oriented evidence should include a bearing- or machine-level holdout and a model frozen before compound labels are opened. Component discrimination and exact-set recovery should be reported together. This contract keeps negative results interpretable and prevents development scores from being presented as deployment guarantees.

6. Conclusion

This study introduced the Sealed Compound Generalization Protocol (SCGP) to separate development, physical-unit-based model selection, and controlled confirmation in rotating machinery fault-diagnosis evaluation. Across the crossed HANOI HUST audit, apparent performance depended materially on partitioning, endpoint aggregation, and whether selection occurred within bearing-grouped outer folds. The model identity also changed across folds, indicating that no candidate was uniformly preferred under physical-unit separation.

Most importantly, strong component discrimination during development did not ensure exact compound recovery at the confirmation boundary. In the disjoint 12-code Paderborn extension, the frozen reference achieved inner-/outer-race AUROC of 0.344/0.719 and exact-set accuracy of 3/12. All four outer-race codes were missed at the joint endpoint. This negative cross-rig result identifies external exact-set recovery as the key validation gap.

SCGP makes the evidential status of each score explicit by recording physical units, aggregation rules, selection permissions, and confirmation-label opening. The present results support reporting source-domain scores as protocol-bound references for deployment-oriented validation. Such studies should retain independent machine-level holdouts, freeze model choice before confirmation, and report component discrimination together with exact-set recovery. Future validation should include more independent units, a common candidate grid, and prospectively registered compound-fault boundaries.

Supplementary Materials: The supporting information contains the extended access-state contract, primary and secondary audit tables, legacy comparator statistics, external boundary details, and reproduction entry points. Supplementary Data S1 contains the manuscript-specific code, feature caches, split manifests, prediction arrays, result files, and audit records.

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Institutional Review Board Statement

Not applicable. The study reanalyzes publicly available machinery benchmark data and involves no human participants, animals, or newly collected personal data.

Informed Consent Statement

Not applicable.

Data Availability Statement

The raw HANOI HUST data are publicly available from Mendeley Data at <https://doi.org/10.17632/cbv7jyx4p9.3> [2]. The Paderborn data are available under CC BY-NC 4.0 from the official Bearing DataCenter [45,46] and are not redistributed here. No new raw data were collected. The manuscript-specific derived feature caches, split manifests, prediction arrays, audit reports, and source code are archived in the SCGP reproducibility package on Zenodo (version 2.1.0; <https://doi.org/10.5281/zenodo.21754415>). The development repository is maintained at <https://github.com/niubisile-wq/HANOI-HUST-SCGP>, and the release materials are also provided with this submission as Supplementary Data S1. The package records the separate three-code pilot and pre-access-frozen 12-code extension registries while preserving the earlier unscored 15-code registry as historical provenance.

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